

Predicting Employee Turnover: Understanding Why People Leave

Employee turnover can cost organizations between 93-200% of each leaver's salary. Understanding why employees leave voluntarily is crucial for organizations to reduce these substantial costs and improve retention. This presentation explores how to analyze, measure, and predict employee turnover at both individual and team levels.

Your future.
Our focus.

Agenda

Understanding Turnover

Definition, importance, and costs of employee turnover

Measuring Turnover

Methods for measuring turnover at individual and team levels

Analyzing Turnover

Statistical approaches to explore differences in turnover patterns

Predicting Turnover

Models to predict who will leave and when they might leave

Business Case

Modeling the costs of turnover and making the case for action

The High Cost of Employee Turnover

The cost of employee turnover can be substantial, projected at 93–200% of each leaver's salary depending on their:

- Skill level
- Level of responsibility
- Difficulty of replacement

For some roles, it can take up to **12 months** for a new joiner to reach the productivity level of their predecessor.

93-200%

Salary Cost

Percentage of salary required to replace an employee

12

Months

Time to reach full productivity for complex roles

Optimize
your workforce,
Minimize costs,

Components of Turnover Costs

1

Separation Costs

- Exit interviews
- Administrative functions
- Severance pay

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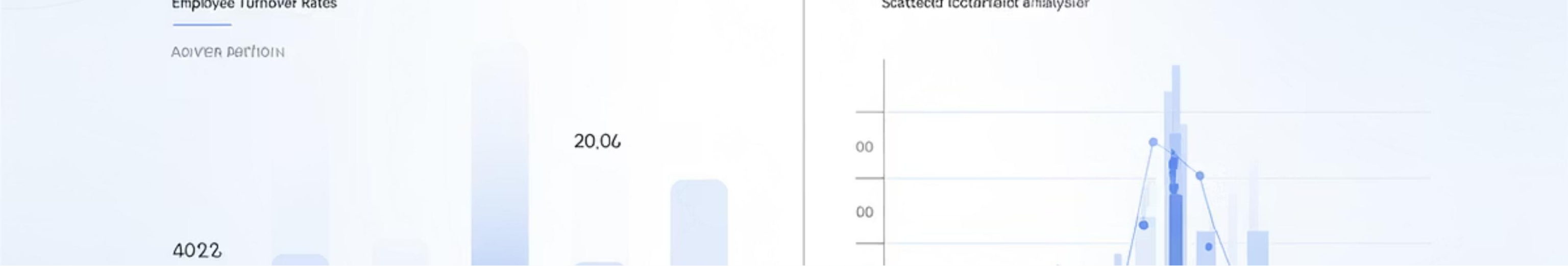
Replacement Costs

- Advertising job openings
- Screening applicants
- Interviewing candidates
- Pre-employment testing

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Training Costs

- Orientation
- Formal training
- On-the-job training
- Reduced productivity during learning curve



Common Approaches to Turnover Analysis

Traditional Approach

Most organizations view turnover reports in spreadsheets showing percentages of people who have left over a given period.

- Simple graphs showing which countries or teams have highest turnover rates
- Little modeling or controlling for explanatory factors
- Often leads to unfounded speculation about causes

Advanced Approach

Using inferential statistical testing across multiple data sources to identify true predictors of turnover.

- Considers multiple possible causes simultaneously
- Controls for confounding variables
- Enables targeted interventions addressing root causes

Measuring Turnover: Organizational Level

The Chartered Institute of Personnel and Development (CIPD) recommends this formula for measuring organizational turnover:

$$\text{Separation Rate} = \frac{\text{Total number of leavers over period}}{\text{Average total number employed over period}} \times 100$$

Other common measures of employee flows include:

Stability Index

Number of staff with 1+ year of service /
Number employed 1 year ago

Survival Rate

Proportion of workers who remain for a
specific time period

Wastage Index

Length of service of leavers / Number
of leavers

Example 1: Exploring Regional Differences in Turnover

Analysis of 1,653 employees across 10 countries in a financial services organization:

Country	Employees	Leavers	Turnover %
Sweden	36	8	22.2%
Australia	47	8	17.0%
Poland	40	5	12.5%
United States	1,067	115	10.8%
Overall	1,653	211	12.8%

Chi-Square Analysis Results

Despite apparent differences in turnover percentages across countries, the chi-square test showed no statistically significant difference ($p = 0.105$).

The current data do not provide sufficient evidence to conclude that turnover rates differ systematically across countries.

Industry average for banking and finance: 8.3% (Mercer, 2023)

Global Turnover Analysis

Q3 2024 Data



Example 2: Team-Level Turnover Analysis

Data Overview

Analysis of 212 teams with 2,910 employees across 4 countries:

- UK: 54 teams
- United States: 73 teams
- Canada: 32 teams
- Spain: 53 teams

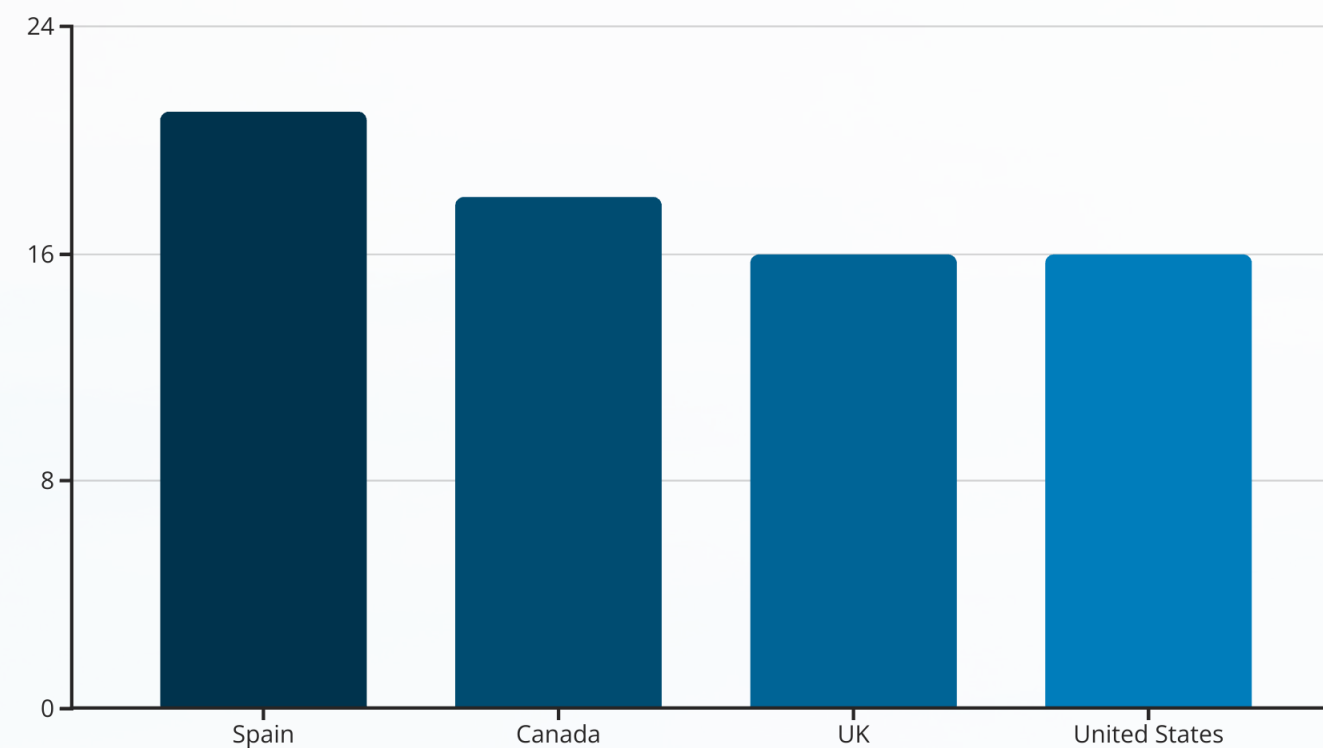
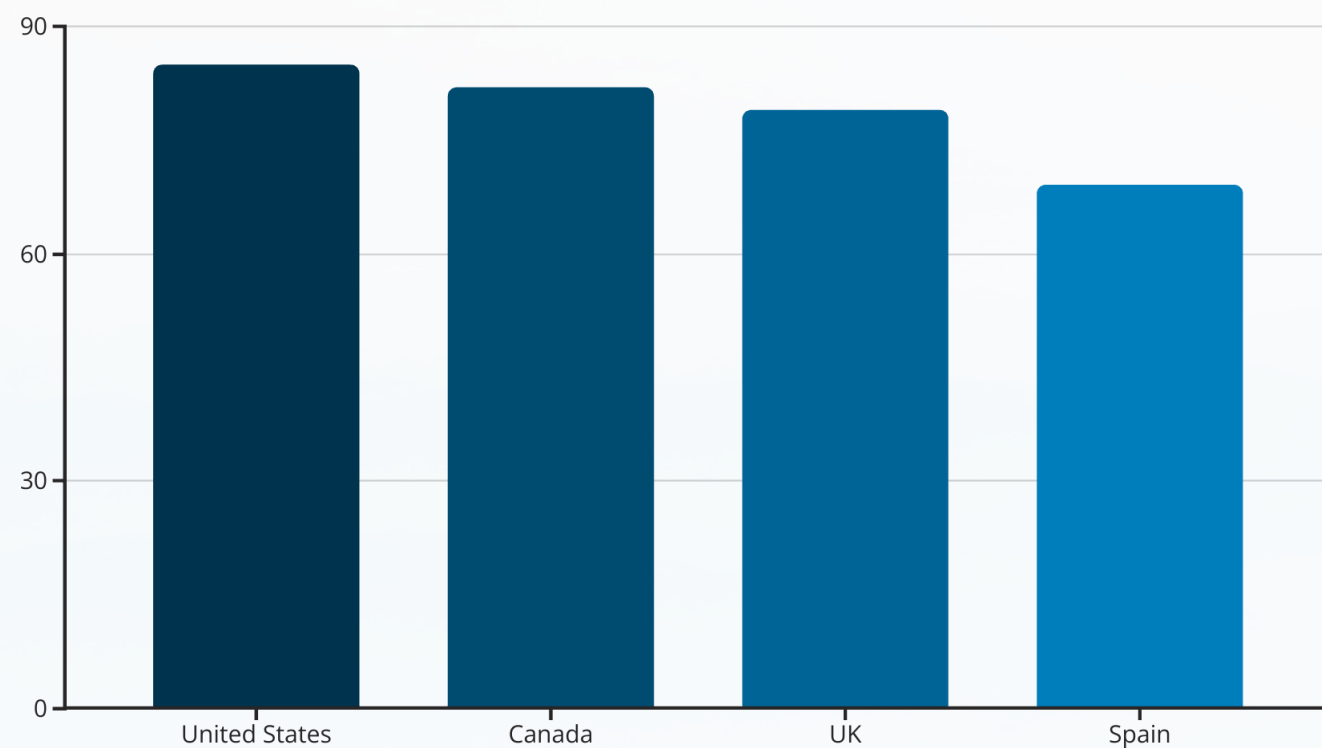
Variables included: team engagement scores, leadership ratings, and other survey data.

Key Findings

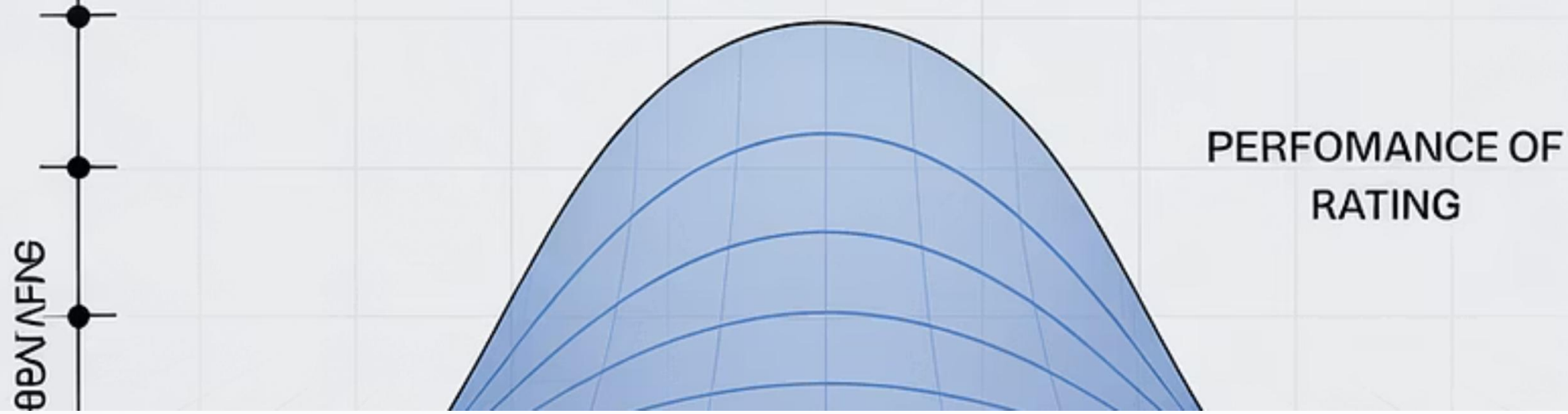
One-way ANOVA analysis revealed:

- Significant differences in team engagement across countries ($p < 0.001$)
- Significant differences in team separation rates across countries ($p < 0.05$)
- Spain had significantly higher turnover than both UK and US
- Spain also had significantly lower engagement scores

Team Engagement and Turnover by Country



Post-hoc tests confirmed significant differences between Spain and other countries for both engagement and turnover rates.



Predicting Individual Turnover: Logistic Regression

Why Logistic Regression?

Individual turnover is a binary event (leaver vs. stayer), making logistic regression the appropriate statistical method.

Logistic regression estimates the probability of an event occurring rather than predicting a precise numerical value.

Model Results

The model including gender, age, length of service, appraisal rating, and country explained 5.6% of turnover variance (Nagelkerke $R^2 = 0.056$).

Two significant predictors emerged:

- **Gender:** Women were 1.84 times more likely to leave than men ($p < 0.001$)
- **Appraisal Rating:** Higher ratings associated with lower turnover ($p < 0.05$)

Understanding Odds Ratios in Turnover Prediction

What is an Odds Ratio?

The odds ratio represents the likelihood of an event occurring (leaving) versus not occurring (staying) when an independent variable increases by one unit.

Gender Effect

Odds ratio = 0.544 for men

This means women are 1.84 times ($1/0.544$) more likely to leave than men, after controlling for other factors.

Appraisal Rating Effect

Odds ratio = 0.780 per rating point

Employees are 28.2% ($1/0.780 = 1.282$) more likely to stay for each one-point increase in their performance rating.



Survival Analysis: How Long Do Employees Stay?

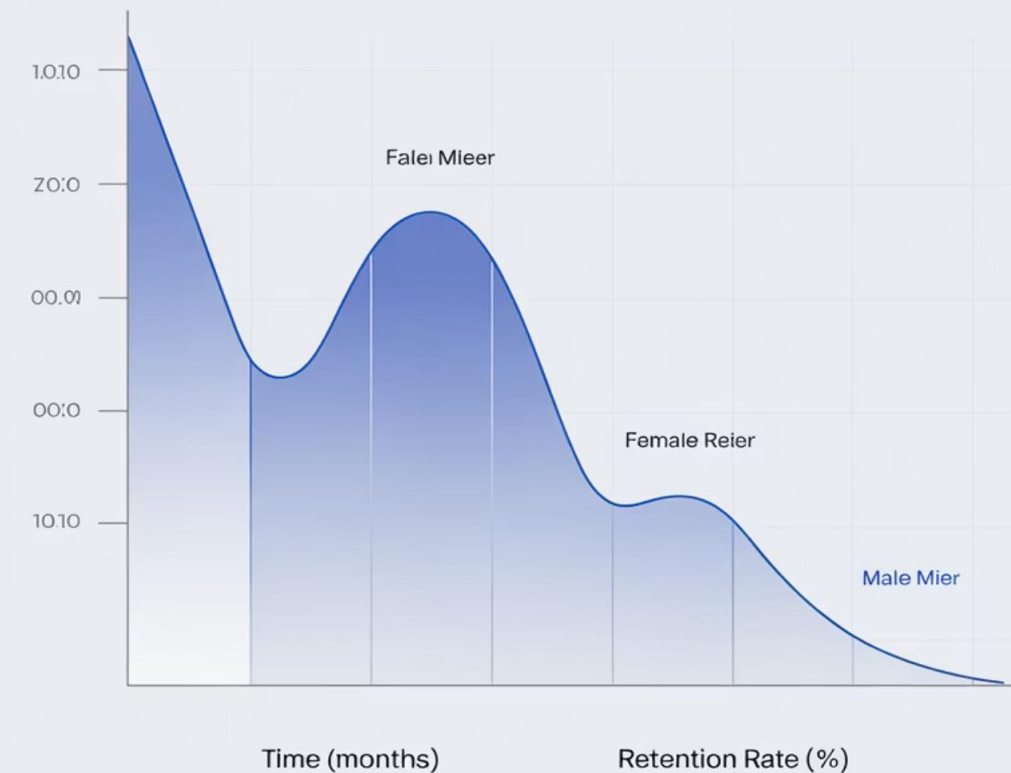
Kaplan-Meier Survival Analysis

This technique helps predict how long employees are likely to remain with the organization before leaving.

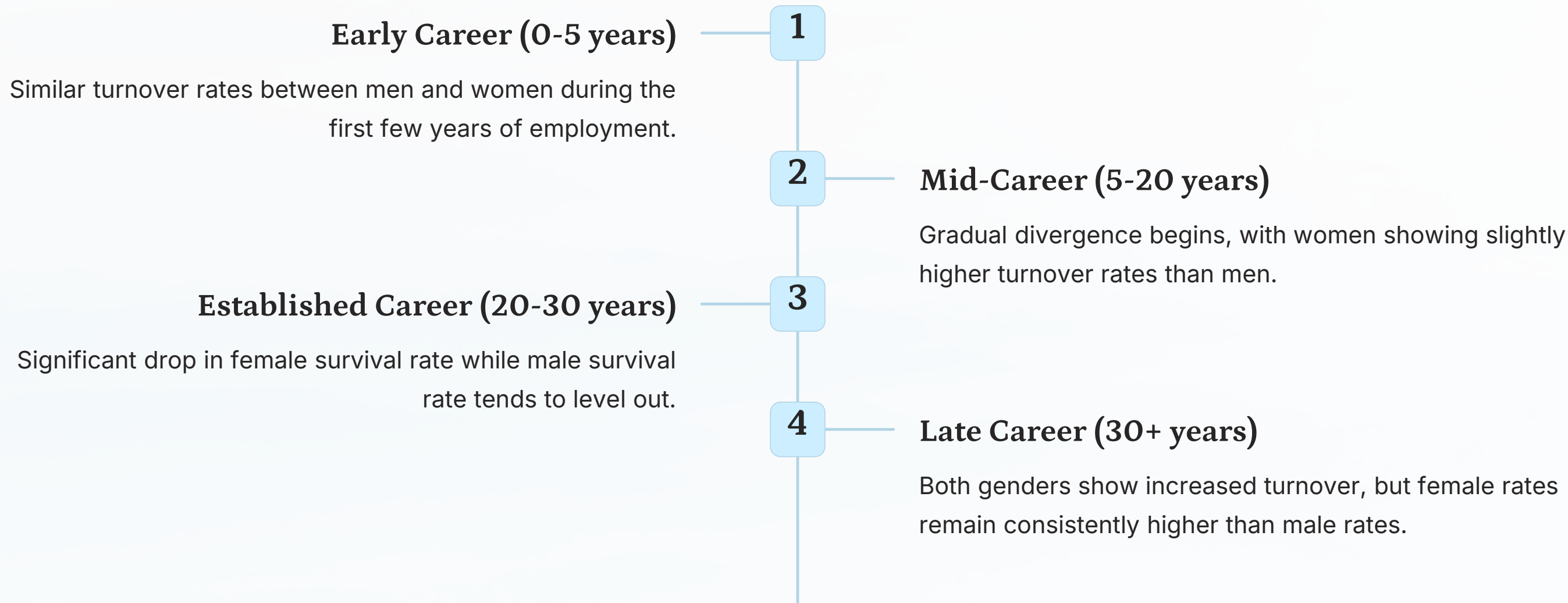
Key findings from our analysis (example):

- Mean survival time for women: 29.1 years
- Mean survival time for men: 35.9 years
- Significant difference between genders ($p < 0.001$)

“Employee Retention Analysis”



Gender Differences in Turnover Patterns (example)



These patterns suggest different career trajectories and potential organizational factors affecting men and women differently over time.



Predicting Team Turnover: Multiple Linear Regression

Model Overview

Multiple linear regression was used because team turnover is a continuous variable (percentage of team members who left).

The model included:

- Six employee attitude/engagement measures
- Team size
- Country (using dummy variables)

Results

The model explained 26.5% of variance in team turnover ($R^2 = 0.265$, $p < 0.001$).

Two significant predictors emerged:

- **Team Engagement:** Higher engagement associated with lower turnover ($p < 0.001$)
- **Drive for Performance:** Higher perception of performance culture associated with higher turnover ($p < 0.05$)

Balancing Performance Culture and Retention

High Performance Expectations

Setting ambitious goals and standards for team performance

Turnover Decision

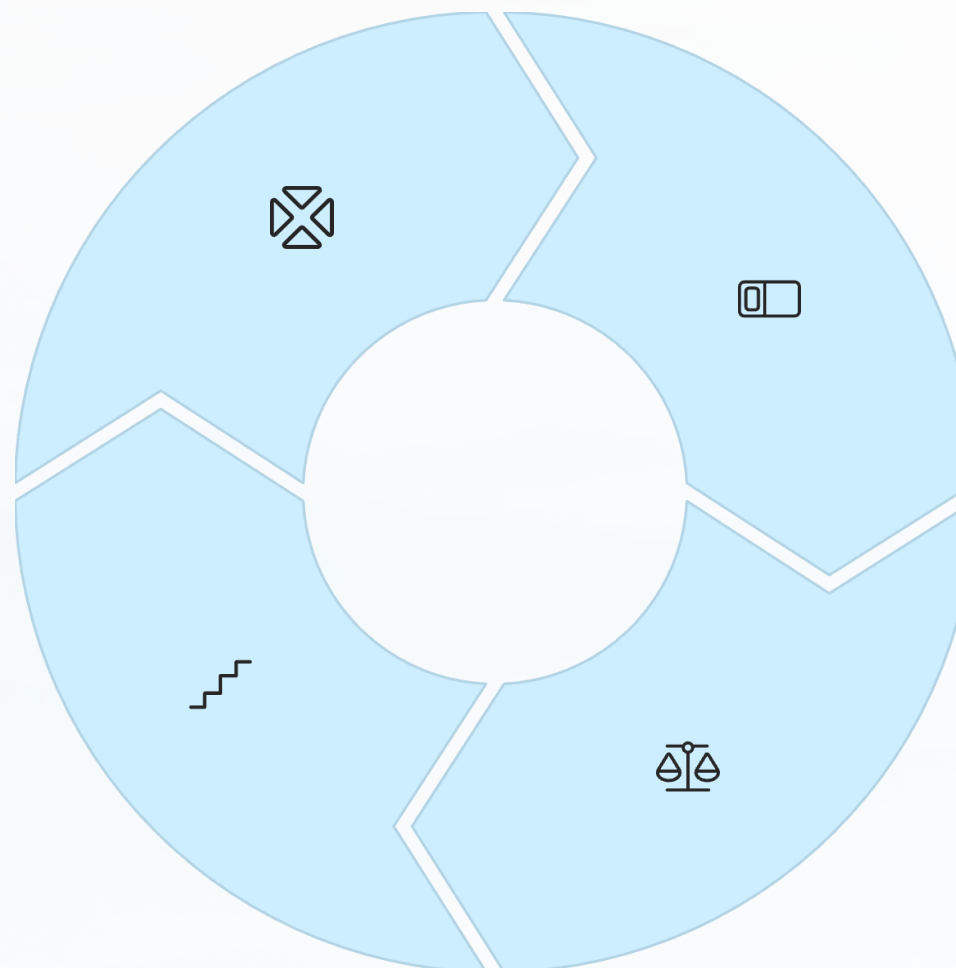
Employees seek opportunities with better balance

Increased Pressure

Employees experience higher stress and workload demands

Work-Life Imbalance

Personal life sacrificed to meet performance expectations



The analysis suggests that an excessive focus on "drive for performance" culture may contribute to higher turnover rates. Organizations should consider how to maintain high performance while supporting employee wellbeing.

A woman with blonde hair tied back, wearing a grey business suit, is seated at a desk in a modern office. She is looking at a tablet computer that displays a dashboard with various charts, including a pie chart and a bar chart. The title on the tablet reads "ROI Calculations for Employee Retention Program". In the background, other office workers are blurred, and the lighting is soft and professional.

Building a Business Case for Turnover Intervention

Quantifying the Gender Effect

Gender alone accounts for approximately 2.4% of turnover variance in our organization.

If total turnover costs are £4 million annually, addressing gender-related factors could potentially save:

$£4,000,000 \times 0.024 = \text{£}96,000$
annually

More Robust Modeling

When controlling for performance ratings, gender uniquely accounts for 1.5% of turnover variance.

This translates to potential savings of:

$£4,000,000 \times 0.015 = \text{£}60,000$
annually

This provides a clear financial justification for targeted retention initiatives focused on gender equity.

Potential Interventions Based on Analysis

Gender Equity Initiatives

- Review promotion and compensation practices for gender bias
- Implement flexible work arrangements
- Develop mentorship programs for women
- Address potential bias in performance evaluation systems

Performance Management Improvements

- Enhance feedback quality and frequency
- Provide development opportunities for lower performers
- Recognize and reward improvement
- Train managers on effective performance conversations

Country-Specific Strategies

- Investigate Spain's lower engagement and higher turnover
- Implement targeted engagement initiatives
- Address cultural factors affecting retention
- Share best practices from high-retention countries

Limitations and Considerations

Statistical Limitations

- Our models explain only a portion of turnover variance (5.6% for individual, 26.5% for team)
- Some country samples had small populations
- Analysis focused only on voluntary turnover
- Cross-sectional rather than longitudinal data

Additional Factors to Consider

- External market conditions and competition
- Industry-specific trends
- Organizational changes and restructuring
- Compensation relative to market
- Career advancement opportunities
- Retirement patterns

Remember that some voluntary turnover may be functional - the goal is not necessarily zero turnover but rather retaining valuable talent.



Advanced Analysis Opportunities



Interaction Effects

Explore how variables interact, such as whether performance ratings affect turnover differently for men versus women.



Cox Regression

Conduct Cox Regression survival analysis to better understand the impact of age and country on length of service.



Segmentation

Separate analysis by retirement, resignation, or other termination reasons to gain more nuanced insights.



Exit Interviews

Combine quantitative analysis with qualitative data from structured exit interviews to identify additional factors.

Key Takeaways

1 Move Beyond Basic Reporting

Simple turnover percentages can be misleading. Statistical testing helps identify true patterns versus random fluctuations.

3 Quantify Financial Impact

Translate statistical findings into potential cost savings to build compelling business cases for retention initiatives.

2 Focus on Significant Predictors

Our analysis identified gender, performance ratings, team engagement, and performance culture as key turnover drivers...

4 Implement Targeted Interventions

Use data-driven insights to design specific interventions addressing root causes rather than symptoms of turnover.

By applying these predictive HR analytics approaches, organizations can significantly reduce turnover costs while improving employee retention and satisfaction.